

Ut:4

-9M

- Suppose you are given a dataset containing information about whether emails are spam or not spam, along with two features:

- the presence of the word "offer" (1: present, 0: absent)
- the presence of the word "free" (1: present, 0: absent)

You are tasked with classifying a new email with the following features values: "offer" = 1 & "free" = 1.

Given the training dataset:

Email	offer	free	Spam
1	1	0	No
2	0	1	yes
3	1	1	yes
4	0	1	No
5	1	1	yes

Calculate probability that the new email is spam using Naive Bayes.

→

i. Given:

Total emails: 5

Spam = yes : 3

Spam = No : 2

ii. Prior Probabilities:

$$P(\text{spam}) = \frac{3}{5}$$

$$P(\text{Not spam}) = \frac{2}{5}$$



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iii. Calculate likelihoods/conditional probabilities:

a. for "spam = yes" (3 mails):

- "offer" = 1, occurs in 2 mails (3 & 5)

- "free" = 1, occurs in 3 mails (2, 3, & 5)

$$\therefore P(\text{offer} = 1 / \text{spam} = \text{yes}) = 2/3$$

$$P(\text{free} = 1 / \text{spam} = \text{yes}) = 3/3 = 1$$

b. for spam = No (2 mails):

- Offer = 1, occurs in 1 out of 2 mails.

- free = 1, occurs in 1 out of 2 mails.

$$\therefore P(\text{offer} = 1 / \text{spam} = \text{No}) = 1/2$$

$$P(\text{free} = 1 / \text{spam} = \text{No}) = 1/2$$

iv. Apply Naïve Bayes:

$$\text{a. } P(\text{spam} = \text{yes} / \text{offer} = 1, \text{free} = 1) \propto P(\text{spam}) \cdot P(\text{offer} = 1 / \text{spam} = \text{yes}) \cdot P(\text{free} = 1 / \text{spam} = \text{yes})$$

$$= \frac{3}{5} \cdot \frac{2}{3} \cdot 1 = \frac{2}{5}$$

$$b. P(\text{Spam} = \text{No} / \text{Offer} = 1, \text{Free} = 1) \propto P(\text{Not spam}) \cdot P(\text{Offer} = 1 / \text{Spam} = \text{No}) \cdot P(\text{Free} = 1 / \text{Spam} = \text{No})$$

$$= \frac{2}{5} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{10}$$

v. Normalize:

$$- \text{Normalizing constant} = \frac{2}{5} + \frac{1}{10} = \frac{1}{2} = 0.5$$

- Now, divide each by constant to get final probability:

$$P(\text{Spam}) = \frac{2/5}{0.5} = \frac{4}{5} = 0.8$$

$$P(\text{Not spam}) = \frac{1/10}{0.5} = \frac{0.1}{0.5} = 0.2$$

Ans:

The probability that the new email with Offer=1 & Free=1, is Spam is 0.8 (80%).

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- Suppose you have the following dataset containing the coordinates of points in a 2-dimensional space.

Point	X-coordinate	Y-coordinate
A	2	3
B	4	7
C	3	5
D	6	9
E	8	6
F	7	8

perform Kmeans clustering.
Assume initial centroid:
(2, 3) & (8, 6).



i. Initial centroids:

$$C_1 : (2, 3)$$

$$C_2 : (8, 6)$$

Iteration 1: (Assign to nearest cluster)

Point	Coordinates	Distance to x_1, y_1 x_2, y_2		Cluster
		$C_1 (2, 3)$	$C_2 (8, 6)$	
A	2, 3	0.00	6.70	C_1
B	4, 7	4.47	4.12	C_2
C	3, 5	2.23	5.09	C_1
D	6, 9	7.21	3.60	C_2
E	8, 6	6.70	0.00	C_2
F	7, 8	7.07	2.23	C_2

Find distance using euclidean distance:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

ex: point A (2, 3) & C_1 (2, 3)

$$\text{distance} = \sqrt{(2-2)^2 + (3-3)^2} = 0.00$$

point A (2, 3) & C_2 (8, 6)

$$d = \sqrt{(8-2)^2 + (6-3)^2} = 6.70$$

Recompute Centroids:

Cluster 1 (C_1): Points A(2, 3), C(3, 5)

$$\therefore \text{New centroid} : \left(\frac{2+3}{2}, \frac{3+5}{2} \right) = (2.5, 4)$$

Cluster 2 (C_2): Points B(4, 7), D(6, 9), E(8, 6), F(7, 8)

$$\therefore \text{New centroid} : \left(\frac{4+6+8+7}{4}, \frac{7+9+6+8}{4} \right) = (6.25, 7.5)$$

Iteration 2:

Point	Coordinates	Distance to $C_1(2.5, 4)$	Distance to $C_2(6.25, 7.5)$	New cluster
A	2, 3	1.11	6.18	1
B	4, 7	3.35	2.30	2
C	3, 5	1.11	4.10	1
D	6, 9	6.10	1.52	2
E	8, 6	5.85	2.30	2
F	7, 8	6.02	0.90	2

Centroids remain the same as in iteration 1
 $\rightarrow$ Convergence reached.

$\therefore$ Final Clusters & centroids.

Cluster 1 centroid (C_1): (2.5, 4)

Cluster 2 centroid (C_2): (6.25, 7.5)

with assignments as:

Cluster 1: A, C

Cluster 2: B, D, E, F

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- Calculate the support & confidence value for all the possible item sets.

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Transaction ID	Items Bought
1	Onion, Potato, Cold drink
2	Onion, Burger, Cold drink
3	Eggs, Onion, Cold drink
4	Potato, Milk, Eggs
5	Potato, Burger, Cold drink, Milk, Eggs

→

- i. List all items & count / Itemset support count

Item	Count (C)	Support = $C / \text{Total transactions}$
Onion	3	$3/5 = 60\%$
Potato	3	$3/5 = 0.6 = 60\%$
Cold drink	4	$4/5 = 0.8 = 80\%$
Burger	2	$2/5 = 0.4 = 40\%$
Eggs	3	$3/5 = 0.6 = 60\%$
Milk	2	$2/5 = 0.4 = 40\%$

ii. 2-itemset Support Count

itemset	Count	Support
{Onion, Potato}	1	$\frac{1}{5} = 20\%$
{Onion, Colddrink}	3	60%
{Onion, Burger}	1	20%
{Onion, Eggs}	1	20%
{Potato, Colddrink}	2	40%
{Potato, Burger}	1	20%
{Potato, Eggs}	2	40%
{Potato, Milk}	2	40%
{Potato, Eggs}		
{Coldrink, Burger}	2	40%
{Coldrink, Eggs}	2	40%
{Coldrink, Milk}	1	20%
{Burger, Eggs}	1	20%
{Burger, Milk}	1	20%
{Milk, Eggs}	2	40%

iii. Confidence Calculations:

Formula:

$$\text{Confidence}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

Rule	Support(A ∪ B)	Support(A)	Confidence
✓ Onion → Cold drink	3	3	100%
Cold drink → Onion	3	4	75%
✓ Potato → Cold drink	2	3	66.67%
✓ Cold drink → Potato	2	4	50%
Potato → Eggs	2	3	66.67%
Eggs → Potato	2	3	66.67%
Potato → Milk	2	3	66.67%
Milk → Potato	2	2	100%
✓ Cold drink → Burger	2	4	50%
✓ Burger → cold drink	2	2	100%
Cold drink → Eggs	2	4	50%
Eggs → Cold drink	2	3	66.67%
✓ Milk → Eggs	2	2	100%
Eggs → Milk	2	3	66.67%
Onion → Potato	1	3	33.34%

Ques

Suppose that the given data the task is to cluster points (with (x, y) representing location) into three clusters, where the points are:

$A_1(2, 10), A_2(2, 5), A_3(8, 4)$,

$B_1(5, 8), B_2(7, 5), B_3(6, 4)$,

$C_1(1, 2), C_2(4, 9)$.

The distance function is Euclidean distance. Suppose initially we assign A_1, B_1 , and C_1 as the center of each cluster, respectively. Use the Kmeans algorithm to show only the 3 clusters centers after the first round of execution with steps.

Initial Clusters: (centroids)

i. Cluster 1 (C_1): $A_1(2, 10)$

ii. Cluster 2 (C_2): $B_1(5, 8)$

iii. Cluster 3 (C_3): $C_1(1, 2)$

Iteration 1:

Distance to

Point	Coordinates	$C_1(2, 10)$	$C_2(5, 8)$	$C_3(1, 2)$	Cluster
A_1	2, 10	0	3.61	8.06	C_1
A_2	2, 5	5.0	4.24	3.16	C_3
A_3	8, 4	8.49	5.0	7.28	C_2
B_1	5, 8	3.61	0	7.21	C_2
B_2	7, 5	7.07	3.61	6.08	C_2
B_3	6, 4	7.21	4.12	5.39	C_2
C_1	1, 2	8.06	7.21	0	C_3
C_2	4, 9	2.24	1.41	7.07	C_2

Q Compute centroids.

Cluster 1: $C_1 : A_1(2, 10)$

$\therefore$ New Centroid: $(2, 10)$

Cluster 2: $C_2 : A_3(8, 4), B_1(5, 8), B_2(7, 5), B_3(6, 4), C_2(4, 9)$

$\therefore$ New Centroid: $\left(\frac{8+5+7+6+4}{5}, \frac{4+8+5+4+9}{5} \right)$
 $= (6.6)$

Cluster 3: $C_3 : A_2(2, 5), C_1(1, 2)$

$\therefore$ New Centroid: $\left(\frac{2+1}{2}, \frac{5+2}{2} \right) = (1.5, 3.5)$

$\therefore$ After 1st round of execution,

3 cluster centers are:

$C_1(2, 10)$

$C_2(6.6)$

$C_3(1.5, 3.5)$

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- Given the confusion matrix, calculate accuracy, Recall, Error rate with description on heart attack risk.

Predicted classes

Actual classes	Classes	Heart Attack Risk - Yes	Heart Attack Risk No
	Heart Attack Risk - Yes	80	220
	Heart Attack Risk - No	150	9500

Confusion Matrix

Actual \ Predicted	Yes (Risk)	No (Risk)	Total
Yes (Risk)	80 (TP)	220 (FN)	300
No (Risk)	150 (FP)	9500 (TN)	9650
Total	230	9720	9950

- TP: 80 (correctly predicted "yes" for heart attack risk)
- FN: 220 (Incorrectly predicted "No" when actual was "yes")
- FP: 150 (Incorrectly predicted "yes" when actual was "No")
- TN: 9500 (Correctly predicted "No" for No risk).

i. Accuracy: Measures correctness of the model.

$$\text{Accuracy} = \frac{TP + TN}{\text{Total}} = \frac{80 + 9500}{9950} = \frac{9580}{9950}$$

$$= 0.9628 \approx 96.28\%$$

The model is 96.28% correct across both classes.

ii. Precision:

Measures how many predicted "yes" cases were actually ~~true~~ correct.

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{80}{80 + 150} = \frac{80}{230}$$

$$\approx 0.3478 \approx 34.78\%$$

Only 34.78% of predicted "yes" cases truly had heart attack risk.

iii. Recall: how many actual yes cases ~~are~~ correctly predicted.

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{80}{80 + 220} = \frac{80}{300}$$

$$\approx 0.2667 = 26.67\%$$

The model captures only 26.67% of actual heart attack cases.

iv. Error Rate: Measures overall incorrect predictions.

$$\begin{aligned} \text{Error Rate} &= 1 - \text{Accuracy} = 1 - 0.9628 \\ &= 0.0372 = 3.72\% \end{aligned}$$

3.72% of predictions are wrong.

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Given the confusion matrix. Calculate Accuracy, Precision, Recall, Error Rate with description on Diabetic Risk.

Classes		Predicted classes	
		Diabetic Risk-Yes	Diabetic Risk-No
Actual classes	Diabetic Risk (Yes)	90	210
	Diabetic Risk (No)	140	9560

→
Confusion Matrix

Actual \ Predicted	Yes (Risk)	No (Risk)	Total
Yes (Risk)	90 (TP)	210 (FN)	300
No (Risk)	140 (FP)	9560 (TN)	9700
Total	230	9770	10,000

i. Accuracy : $\frac{TP+TN}{Total} = \frac{90+9560}{10,000} = 0.965 = 96.5\%$

ii. Precision : $\frac{TP}{TP+FP} = \frac{90}{90+140} \approx 0.3913 = 39.13\%$

iii. Recall : $\frac{TP}{TP+FN} = \frac{90}{90+210} = 0.3 = 30\%$

iv. Error Rate : $1 - \text{Accuracy} = 1 - 0.965 = 0.035 = 3.5\%$